PART 1 : PHYSICS

SECTION-I (i)

1) Ra1 of mass 50 kg was moving with initial velocity of $20\hat{i} + 30\hat{j} + 40\hat{k}$. G1 applied a net external force on him and the value of force was $100\hat{i} + 200\hat{j} + 300\hat{k}$ for 10s. Assume Ra1 to offer no resistance, the angle made by his velocity vector after 10s with x-axis will be :-

(A) $\cos^{-1}\frac{7}{\sqrt{165}}$

(B) $\cos^{-1}\frac{10}{\sqrt{165}}$

(C) $\cos^{-1}\frac{4}{\sqrt{165}}$

(D) 0°

2)

The dimensional formula for ε_0 is :

(A) $M^{-2}L^{-3}I^{3}T^{4}$

(B) $M^{-1}L^{-3}I^{2}T^{4}$

(C) $M^{-1}L^{-3}I^{3}T^{2}$

(D) $M^{-2}L^{-3}I^{2}T^{2}$

3) A charged particle of mass m = 2 kg and charge 1μC is projected from a horizontal ground at an angle θ = 45° with speed 10 ms^{-1}. In space, a horizontal electric field E = 2×10^{7} NC^{-1} exists. The range of the projectile is

(A) 20m

(B) 60m

(C) 200m

(D) 180m

4) Two completely different planets X and Y are each composed of the same material. The planet Y has radius twice as large as that of X. The ratio of the gravitational acceleration g_y at the surface of Y to the gravitational acceleration g_x at the surface of X is given by :-

(A) $\frac{1}{8}$

(B) $\frac{1}{2}$

(C) 2

(D) $\frac{1}{4}$

SECTION-I (ii)

1) The value of $\left|\vec{A}+\vec{B}-\vec{C}+\vec{D}\right|$ can be zero if :—

(A) $\left|\vec{A}\right| = 5,\ \left|\vec{B}\right| = 3,\ \left|\vec{C}\right| = 4,\ \left|\vec{D}\right| = 13$

(B) $\left|\vec{A}\right| = 2\sqrt{2},\ \left|\vec{B}\right| = 2,\ \left|\vec{C}\right| = 2,\ \left|\vec{D}\right| = 5$

(C) $\left|\vec{A}\right| = 2\sqrt{2},\ \left|\vec{B}\right| = 2,\ \left|\vec{C}\right| = 2,\ \left|\vec{D}\right| = 10$

(D) $\left|\vec{A}\right| = 5,\ \left|\vec{B}\right| = 4,\ \left|\vec{C}\right| = 3,\ \left|\vec{D}\right| = 8$

2) The angle between two vectors $\vec{a}$ & $\vec{b}$ is θ and the magnitude of $\vec{b}$ is half of magnitude of $\vec{a}$. If $\vec{c} = \vec{a} - \vec{b}$ & $|\vec{a}| = a$ then choose the correct statements

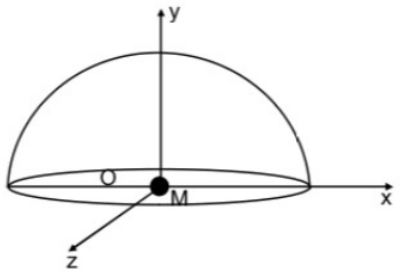
(A) if $c = \frac{a\sqrt{5}}{2}$ then θ will be 90°

(B) if $c = \frac{a\sqrt{3}}{2}$ then θ will be 60°

(C) if $c = \frac{a}{\sqrt{2}}$ then θ will be 45^{0}

(D) if $c = \frac{3a}{2}$ then θ will be 180°

3) A hemispherical shell of mass 2M and radius 6R and a point mass M are performing circular motion due to their mutual gravitational interaction. Their positions are shown in figure at any moment of time during motion. If r_1 and r_2 are the radii of circular path of hemispherical shell and point mass respectively and ω_1 and ω_2 are the angular speeds of hemi-spherical shell and point mass



respectively, then choose the correct statement(s).

(A) $\frac{r_1}{r_2} = \frac{2}{1}$

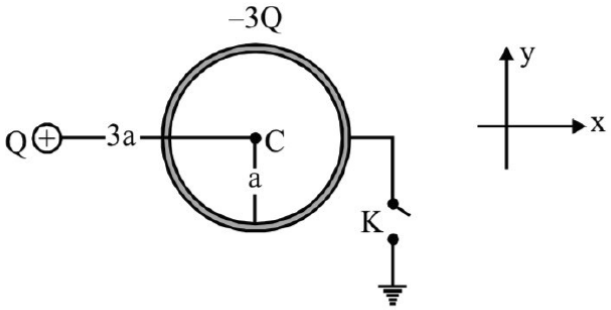
(B) $\frac{r_1}{r_2} = \frac{1}{2}$

(C) $\omega_1 = \sqrt{\frac{GM}{72R^3}}$

(D) $\omega_2 = \sqrt{\frac{GM}{72R^3}}$

4) A point charge +Q is kept at a distance 3a from the centre of a spherical conducting shell of charge -3Q and radius a.

Assuming $K = \frac{1}{4\pi\varepsilon_0}$, mark the **CORRECT** statements(s):-



Before closing switch the electric field at C due to the spherical shell is

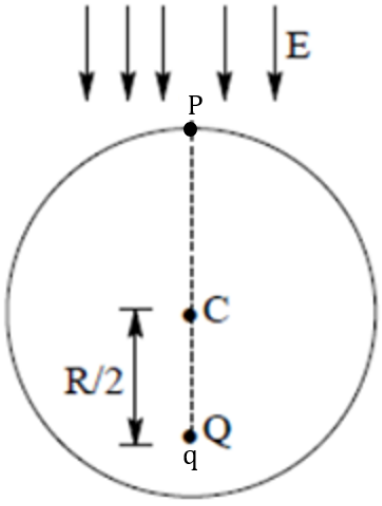
(A) $\left(-\frac{KQ}{9a^2}\hat{i}\right)$

(B) Before closing switch the potential of the shell is $\left(-\frac{8KQ}{3a}\right)$

(C) After closing the key,the charge on the shell is $\left(-\frac{Q}{3}\right)$

(D) After closing the key, the charge of the shell is zero

5) A positive charge q is placed inside a neutral hollow conducting sphere of radius R, as shown in figure. Whole system is placed in uniform external vertical electric field pointing downward (line PCQ is also vertical) then select the correct statement(s) about electric field at point P. Point P is a point of the material inside the conductor.



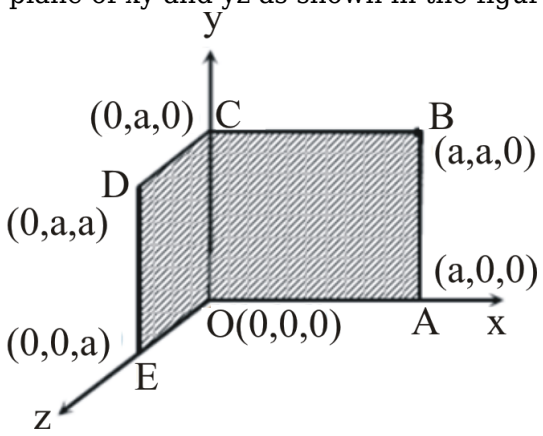
(A) Electric field due to outer surface of sphere at point P is zero

(B) Magnitude of electric field due to inner surface of sphere at point Pis $q/9\pi\varepsilon_0 R^2$

(C) Magnitude of electric field at point P, due to hollow sphere is less than E

(D) Direction of resultant electric field at point P due to charge q and hollow sphere is vertically upward

6) Two surface OABC and OCDE lies in the plane of xy and yz as shown in the figure. A charged



particle 'q' lies in the space at a point P. if

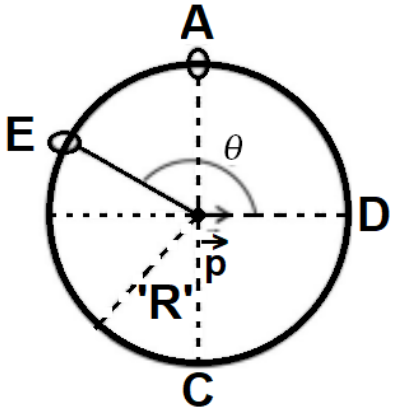
(A) coordinates of 'P' is (a − Δr, a − Δr, Δr) and a >> Δr, then flux passing through surface OABC is $\frac{7q}{24\varepsilon_0}$

(B) coordinates of 'P' is (a − Δr, a − Δr, Δr) and a >> Δr, then flux passing through surface OCDE is $\frac{q}{24\varepsilon_0}$

(C) coordinates of 'P' is (a + Δr, a + Δr, −Δr) and a >> Δr, then flux passing through surface OABC is $\frac{q}{24\varepsilon_0}$

(D) coordinates of 'P' is (a + Δr, a + Δr, −Δr) and a >> Δr, then flux passing through surface OCDE is $\frac{q}{24\varepsilon_0}$

7) A ring of radius R is fixed and kept in a vertical plane. A small dipole of dipole moment p is fixed at the centre of ring. A small bead of mass'm' and charge q is released from point A. Bead can slide on the ring. The inner diameter of bead and thickness of ring are nearly equal. Ring is insulating and frictionless. Ignore effect of gravity. Now choose the correct option(s).



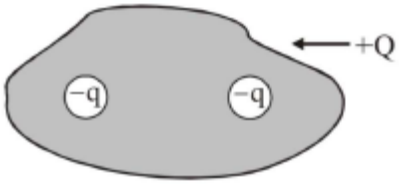
(A) Bead will never reach at point D

(B) The normal force between bead and ring at 'E' is directly proportional to the charge on the bead.

(C) The motion of bead is periodic.

(D) Speed of bead at point 'E' is given $\sqrt{\frac{-p(\cos\theta)q}{2\pi\varepsilon_0 mr^2}}$

8) A conducting body is given charge Q and charge -q has been placed in each of the cavity, which of the following statements is/are true?



(A) If Q = 2q, then conducting body will be at zero potential

(B) If an external electric field is applied then the charge distribution on the outer surface of conductor would change.

(C) The potential of any point inside the cavity is less than that of conducting body.

(D) None of these

SECTION-II

1) The maximum and minimum magnitudes of the resultant of two forces are 7 N and 1 N respectively. Find the magnitude of resultant force when they act perpendicular to each other :-

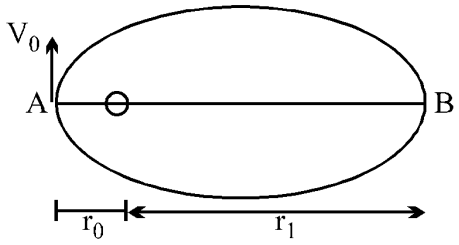
2) Two forces $\vec{F}_1$ and $\vec{F}_2$ have a resultant $\vec{F}_3$. If $\vec{F}_1 = 2\hat{i} - 3\hat{j}$ and $\vec{F}_3 = 5\hat{i} - 4\hat{j}$ then magnitude of $\vec{F}_2$ is X. The value of $\frac{X}{\sqrt{10}}$ is

3) Using dimensional analysis, angular frequency (ω) of surface waves on a lake as a function of surface tension σ, water density Q and the wave number k = 2π/λ (λ is the wavelength) is derived. Assuming that the dimensionless coefficient is equal to two (in SI units) in this case, find the value of group velocity v_g (in m/s) where $v_g = \frac{\partial\omega}{\partial k}$.
(Use : σ = 64mN·m^{-1}, Q = 320π kg·m^{-3}, λ = 1 cm)

4) Two particles of combined mass M, placed in space with certain separation, are released. Interaction between the particles is only of gravitational in nature and there is no external force present. Acceleration of one particle with respect to each other when separation between them is R, has a magnitude $\frac{xGM}{R^2}$. Value of x is

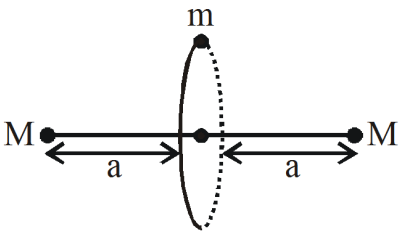
5) A satellite describes an elliptic orbit about a planet. Denoting by r_0 and r_1, the distances corresponding to the perigee and apogee of the orbit . The radius of curvature of the orbit at point A

0



and B in terms of r and r_1 is $\frac{\alpha r_0 r_1}{3(r_0 + r_1)}$. The value of α is.

6) A particle of mass m is moving in circular motion of radius r with constant speed v in influence of two fixed particles of mass M. Plane of circular motion is perpendicular to line joining fixed particles as shown. Speed of particle is given as $v = \sqrt{\frac{GM}{nr}}$, find value of n ($a = r\sqrt{3}$)



PART 2 : CHEMISTRY

SECTION-I (i)

1) 0.7 g of an organic compound on heating evolved all of its nitrogen into ammonia which is reacted with 200 ml of 0.25 M HNO_3 solution. The residual HNO_3 solution is reacted completely with 30 ml of 0.5 M $Ca(OH)_2$. % by mass of nitrogen in the organic compound is :

(A) 20

(B) 15

(C) 30

(D) 40

2) A person adds 1.71 gram of sugar ($C_{12}H_{22}O_{11}$) in order to sweeten his tea. The number of carbon atoms added are
[Given : Molecular mass of sugar = 342,$N_A = 6 \times 10^{23}$]

(A) 3.6×10^{22}

(B) 7.2×10^{21}

(C) 0.05

(D) 6.6×10^{22}

3) What volume of 0.1 M $KMnO_4$ solution is required to completely oxidise 144 mg of FeC_2O_4 in acidic medium ?
(Mol.wt of Ferrour oxalate = 144 g/mole)

$FeC_2O_4 + MnO_4^- + H^+ \rightarrow Fe^{3+} + CO_2 + Mn^{2+} + H_2O$

(A) 2 ml

(B) 4 ml

(C) 6 ml

(D) 8 ml

4) One mole of acidified $K_2Cr_2O_7$ on reaction with sufficient KI will liberate ..... mole(s) of I_2, along with $Cr^{3\oplus}$.

(A) 6

(B) 1

(C) 7

(D) 3

SECTION-I (ii)

1) Which of the following options is/are not correct w.r.t oxidation number (O.N.) ?

(A) The oxidation state of both the Nitrogen atoms in NH_4NO_3 is +1.

(B) O.N. of Sulphur in FeS_2 is –2.

(C) O.N. of F can never be positive.

(D) O.N. of alkaline earth metals can either be 0 or +2 and no other value.

2) When a equimolar mixture of Cu_2S and CuS is titrated with $Ba(MnO_4)_2$ in acidic medium, the final products contain Cu^{2+}, SO_2 and Mn^{2+}. If the Mol. mass of Cu_2S, CuS and $Ba(MnO_4)_2$ are M_1, M_2 and M_3 respectively then :

(A) Eq. mass of Cu_2S is $\frac{M_1}{8}$

(B) Eq. mass of CuS is $\frac{M_2}{6}$

(C) Eq. mass of $Ba(MnO_4)_2$ is $\frac{M_3}{5}$

(D) Cu_2S and CuS both have same equivalents in mixture.

3) Choose the incorrect statement(s)

(A) 1 gm-molecule always contains same number of atoms

(B) weight of one molecule in gm is equal to its molar mass

(C) Number of atoms in 2 gm of hydrogen is greater than 11.35 litre hydrogen at STP

(D) One mole of oxygen and one mole of helium contain same number of atoms at STP.

4) Lithium metal reacts with nitrogen gas to produce a white solid Lithium nitride (Li_3N) according to the reaction:
(Atomic mass of Li = 7, N = 14)

$6\ Li\ (s) + N_2\ (g) \rightarrow 2Li_3N(s)$

If 8.4 g of Li is taken initially with excess of N_2 (g) then :

(A) Volume of nitrogen gas consumed at NTP is 4.48 L

(B) Total mass of the product obtained is 14g

(C) Total number of atoms obtained in the product is 1.6 N_A

(D) Lithium and nitrogen combine in ratio 3 : 2 by mass.

5) A sample of oxygen contains O^{16} & O^{18} isotopes only with percentage abundance respectively as 90% & 10%.
Identify the correct option(s) :

(A) Average atomic mass of sample is 16.2.

(B) Average number of protons per atom is 8.

(C) Average number of neutrons per atom is 8.2.

(D) Molecular mass of oxygen gas which can be formed from the sample can vary from 32 to 36.

6) The number of hydrogen atoms in 0.9 gm glucose, $C_6H_{12}O_6$, is same as

(A) 0.048 gm hydrazine, N_2H_4

(B) 0.17 gm ammonia, NH_3

(C) 0.30 gm ethane, C_2H_6

(D) 0.03 gm hydrogen, H_2

7) 40 ml, 0.05 M solution of sodium sesquicarbonate dihydrate [$Na_2CO_3.NaHCO_3.2H_2O$] is titrated against 0.05 M HCl solution, x ml of acid is required to reach the phenolphthalein end point, while y ml. of same acid was required when methyl orange indicator was used in a separate titration. Which of the following statements are correct ?

(A) y – x = 80 ml

(B) y + x = 160 ml

(C) If the titration is started with phenolphthalein indicator and methyl orange is added at the end point, 2x ml. HCl would be required further to reach end point.

(D) y + x = 120 ml.

8) The correct statement(s) is/are

(A) The n–factor of HNO_3 in the following reaction is $\frac{4}{5}$
$Zn + HNO_3 \rightarrow Zn(NO_3)_2 + N_2O + H_2O$

(B) The reaction $(NH_4)_2Cr_2O_7 \rightarrow N_2 + Cr_2O_3 + H_2O$ is a intramolecular redox reaction.

(C) The equivalent mass of $(NH_4)_2Cr_2O_7$ in option (B) is $\frac{M}{6}$ if M is molecular mass of $(NH_4)_2Cr_2O_7$.

(D) In the reaction $Na + H_2 \rightarrow NaH,$ H_2 acts as oxidising agent.

SECTION-II

1) What mass (kg) of H_2SO_4 could be produced in the process given below, if 5.52kg of FeS_2 is used.
$FeS_2(s) + O_2(g) \rightarrow Fe_2O_3(s) + SO_2(g)$
$SO_2(g) + O_2(g) \rightarrow SO_3(g)$
$SO_3(g) + H_2O(\ell) \rightarrow H_2SO_4(aq)$

2) Potassium dichromate in acidic medium reacts with H_2S to oxidise it as per given unbalanced equation. How much kg of $K_2Cr_2O_7$ must be required if 1000 kg of sulphur is to be produced by following reaction ? (Molar mass of $K_2Cr_2O_7$ = 294 g/mol).
$$K_2Cr_2O_7 + H_2SO_4 + H_2S \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + S + H_2O$$

3) Mass percentage of oxygen in divalent metal carbonate is 36%. If the number of metal atoms in 266.66 mg of this metal carbonate sample is $x \times 10^{20}$ then find value of x [$N_A = 6.02 \times 10^{23}$]

4) A mineral consists of an equimolar mixture of the carbonates of two bivalent metals. One metal is present to the extent of 13.2% by weight. 2.58 g of the mineral on heating lost 1.232 g of CO_2. Calculate the % by weight of the other metal.

5) The following redox reaction occurs in basic medium : $NO_3^- + Zn(s) \rightarrow Zn^{2+} + NH_4^+$, when the above reaction is balanced such that the stoichiometric coefficients are in smallest whole number ratio, then the difference of stoichiometric coefficient of OH^- and Zn(s) ion will be -

6) Calculate the mass of anhydrous oxalic acid, which can be oxidised to CO_2 (g) by 100 ml of an MnO_4^- solution, 10 ml of which is capable of oxidising 50 ml of 1N I^- to I_2

PART 3 : MATHEMATICS

SECTION-I (i)

1) The number of integral values of x satisfying the inequality $[x-5][x-3] + 2 < [x-5] + 2[x-3]$ (where [.] represents greatest integer function) is

(A) 0
(B) 1
(C) 2
(D) 3

2) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a real valued function such that $f(10 + x) = f(10 - x)\ \forall\ x \in \mathbb{R}$ and $f(20 + x) = -f(20 - x)\ \forall\ x \in \mathbb{R}$. Then which of the following statements is true

(A) f(x) is odd and periodic
(B) f(x) is odd and aperiodic
(C) f(x) is even and periodic
(D) f(x) is even and aperiodic

3) Let $f, g : \mathbb{N} - \{1\} \to \mathbb{N}$ be functions defined by f(a) = α, where α is the maximum of the powers of those primes p such that p^{α} divides a, and $g(a) = a + 1$, for all $a \in \mathbb{N} - \{1\}$. Then, the function f + g is

(A) one-one but not onto

(B) onto but not one-one

(C) both one-one and onto

(D) neither one-one nor onto

4) Out of the members of three athletic teams in a school 21 are in the cricket team, 26 are in the hockey team and 29 are in the football team. Among them, 14 play hockey and cricket, 15 play hockey and football, and 12 play football and cricket. Eight play all the three games. Find the total number of members in the three athletic teams.

(A) 41

(B) 42

(C) 43

(D) 44

SECTION-I (ii)

1) f(x) = $\begin{cases} [x], & -2 \le x \le -1 \\ |x|+1, & -1 < x \le 2 \end{cases}$ and g(x) = $\begin{cases} [x], & -\pi \le x \le 0 \\ \sin x, & 0 \le x \le \pi \end{cases}$, [.] denotes the greatest integer function. If h(x) = g(f(x))

(A) Domain of h(x) is [–1, 1]

(B) Range of h(x) is {–2, –1} ∪ [sin3, 1]

(C) Domain of h(x) is [–2, 2]

(D) Range of h(x) is {–2, –1} ∪ [sin2, 1]

2) Let $f : \{1,2,3,4,5\} \to \{1,2,3,4,5\}$ is such that $f(x)$ is a one-one function satisfying following condition
$f(x) = x + 1$ if and only if x is even (i.e. $f(3) \neq 4$, $f(4) = 5$ etc). Then $f^{-1}(2)$ can be

(A) 1

(B) 3

(C) 5

(D) 2

3) If a function is defined by an implicit equation $2^{|x|+|y|} + 2^{|x|-|y|} = 2$, then

(A) Domain of function is singleton

(B) Range of function is singleton

(C) graph of the function intersects the line y = x

(D) maximum value of function is 2

4) If f is a function defined from R to R such that $f(f(f(x))) - (f(f(x)))^3 = f(f(x))$, then

(A) $f(x)$ has exactly two real roots

(B) $f(x)$ has exactly three real roots

(C) $f(x)$ is surjective function

(D) $f(x) = f^{-1}(x)$ at $x = 0$

5) Let $f : R \to R$ is polynomial satisfying $(x - 4)f(x - 1) = (x - 7)f(x)$, $\forall x \in R$. If $f(7) = 6$, then-

(A) $f'(x)$ will have three distinct positive real roots

(B) $f'(x)$ will have two distinct positive real roots

(C) f is surjective

(D) f is injective

6) Number of integral root(s) of the function $x^8 - 22x^5 - 6x^3 + 3x^2 + 55 = 0$ is 'n', then n can **not** be

(A) 0

(B) 1

(C) 2

(D) 3

7) Let r be a relation from R (Set of real number) to R defined by r = {(x, y) | x, y ∈ R and xy is an irrational number}, then which of the following is/are incorrect

(A) relation is reflexive and symmetric only

(B) relation is symmetric only

(C) relation is symmetric and transitive only

(D) relation is an equivalence relation

8) Let α, β, (α > β) be roots of equation $x^2 - x - 1 = 0$. Define $T(n) = \frac{\alpha^n - \beta^n}{\sqrt{5}}$, $\forall n \in N$, then which of following are **True** ?

(A) T(2023) = T(2022) + T(2021)

(B) T(10) = 55

(C) $\sum_{n=1}^{2021} T(n) = T(2023) - 1$

(D) $\sum_{n=1}^{2021} T(n) = T(2023) + 1$

SECTION-II

1) Let $f(x) = \left[\frac{1}{\cos\{x\}}\right]$ and $g(x) = 2x^3 - 3x(k + 1) + k(3k + 1)$. If $g(f(x)) < 0$ $\forall x \in R$, then number of integral values of k is

(where [.] & {.} represent greatest integer function & fractional part function respectively)

2) Let $f : [-2,2] \to R$, where $g(x) = x^{2015} + \text{sgn}(x) + \left[\frac{x^2+1}{p}\right]$ be an odd function for all $x \in [-2,2]$, then the smallest integral value of p is (where [.] denotes greatest integer function and sgn(.) denotes signum function)

3) The number of integral values of x satisfying equation $\text{sgn}\left(\left[\frac{15}{1+x^2}\right]\right) = [1 + \{2x\}]$ is equal to, here sgn x, [.] and {.} denotes signum, greatest integer and fractional function respectively

4) Let $f(x) = x^3 - x^2 - x + 1$, $g(x) = a^2 - 2$ (where a is real parameter) and f(x) = g(x) has three real and distinct roots, then $f(x) = [a^2]$ has real root having the integral part (where [.] is G.I.F.) equal to

5) Let x, y and z be positive roots and $x^2 + xy + y^2 = 2$, $y^2 + yz + z^2 = 1$ and $z^2 + zx + x^2 = 3$. If value of xy + yz + zx can be expressed as $\sqrt{\frac{\alpha}{\beta}}$ where a and b are relatively prime integers. Then value of $\alpha - \beta$ is

6) If $x_i \in \{-1, 0, 1, 2\}$; i = 1, 2, ..., n and $\sum_{i=1}^{n} x_i = 19$, $\sum_{i=1}^{n} x_i^2 = 99$, then minimum value of $\sum_{i=1}^{n} x_i^3$ is

ANSWER KEYS

PART 1 : PHYSICS

SECTION-I (i)

Q.	1	2	3	4
A.	C	B	A	C

SECTION-I (ii)

Q.	5	6	7	8	9	10	11	12
A.	B,D	A,B,D	B,C,D	A,B,C	B,D	A,B,C,D	A,B,C,D	A,B,C

SECTION-II

Q.	13	14	15	16	17	18
A.	5.00	1.00	0.60	1.00	6.00	4.00

PART 2 : CHEMISTRY

SECTION-I (i)

Q.	19	20	21	22
A.	D	A	C	D

SECTION-I (ii)

Q.	23	24	25	26	27	28	29	30
A.	A,B	A,B	A,B,D	A,B,C,D	A,B,C,D	C	A,B,C	A,B,C,D

SECTION-II

Q.	31	32	33	34	35	36
A.	9.00	3062.50	12.04	21.70	6.00	22.50

PART 3 : MATHEMATICS

SECTION-I (i)

Q.	37	38	39	40
A.	C	A	D	C

SECTION-I (ii)

Q.	41	42	43	44	45	46	47	48
A.	B,C	B,C	A,B,C	C,D	B,C	B,C,D	A,C,D	A,B,C

SECTION-II

Q.	49	50	51	52	53	54
A.	**1.00**	**6.00**	**7.00**	**1.00**	**5.00**	**19.00**

SOLUTIONS

PART 1 : PHYSICS

1)

$F_{net} = 100\hat{i} + 200\hat{j} + 300\hat{k} \Rightarrow \vec{a} = 2\hat{i} + 4\hat{j} + 6\hat{k}$

 velocity of Ra1 after 10 s = $40\hat{i} + 70\hat{j} + 100\hat{k}$

 Angle made by velocity vector with x-axis = $\cos^{-1}\frac{4}{10\sqrt{165}}$

2) $F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$

$[\varepsilon_0] = \left[\frac{q_1 q_2}{4\pi F r^2}\right]$

$$a_x = \frac{qE}{m} = \frac{10^{-6}\times 2\times 10^7}{2} = 10 ms^{-2}$$

$$\text{Time of flight } T = \frac{2u\sin\theta}{g} = \frac{2\times 10\times\frac{1}{\sqrt{2}}}{10}$$

$$T = \sqrt{2}\ \text{sec}$$

$$\text{Hence, range } R = u_x T + \frac{1}{2}a_x T^2$$

$$R = 10\cos 45^\circ \times T + \frac{1}{2}a_x T^2$$

$$= \frac{10}{\sqrt{2}}\sqrt{2} + \frac{1}{2}\times 10\times 2$$

3) $\Rightarrow R = 20m$

4) $g = \frac{GM}{R^2} = \frac{4}{3}G\pi\rho R$

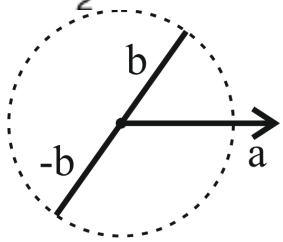
$\frac{g_y}{g_x} = \frac{R_y}{R_x} = \frac{8M_x}{4M_x} = 2$

5) $\left(\vec{A} + \vec{B} - \vec{C} + \vec{D}\right)$ can not be zero if sum of magnitude of three smallest vectors is less than the magnitude of bigger vector.

6) $\vec{a}.\vec{b} = \vec{a} + \left(-\vec{b}\right)$

$\Rightarrow \ |\vec{a}| = 2|\vec{b}| = a$

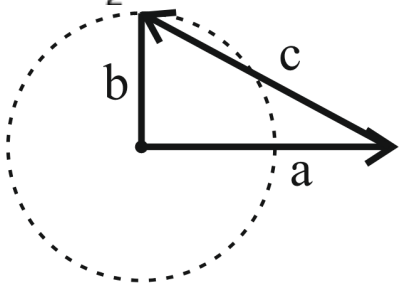
$|b| = \frac{a}{2}$



(A) $|\vec{c}| = \sqrt{|a|^2 + |b|^2 + 2\,|a|\,|b|\cos\theta}$

$\Rightarrow |\vec{c}| = \sqrt{a^2 + \frac{a^2}{4} + a \times a.\cos\theta}$

$|c| = \frac{\sqrt{5}a^2}{2}$



(B) $\frac{a\sqrt{3}}{2} = \sqrt{a^2 + \frac{a^2}{4} + a^2\cos\theta}$

$\Rightarrow \frac{3a^2}{2} = \frac{5a^2}{4} + a^2\cos\theta$

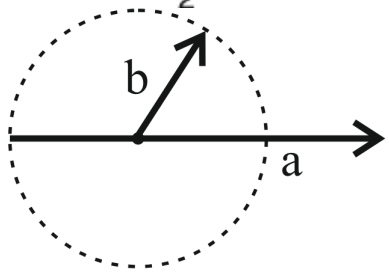
$\Rightarrow \frac{a^2}{4} = a^2\cos\theta$

$\Rightarrow \theta = 60^o$

(C) $\left(\frac{a}{\sqrt{2}}\right)^2 = \frac{5a^2}{4} + a^2\cos\theta$

$\Rightarrow \frac{a^2}{2} - \frac{5a^2}{4} = a^2\cos\theta$

$\Rightarrow \cos\theta = -\frac{\sqrt{3}}{2}$



$\left(\frac{3a}{2}\right)^2 = \frac{5a^2}{4} + a^2\cos\theta$

$\frac{9a^2}{4} - \frac{5a^2}{4} = a^2\cos\theta$

$\Rightarrow \cos\theta = -1$

$\theta = 180^0$

7) The system is equivalent to a binary system as shown in the diagram.

$r_1 = \frac{M \times 3R}{M + 2M} = R$

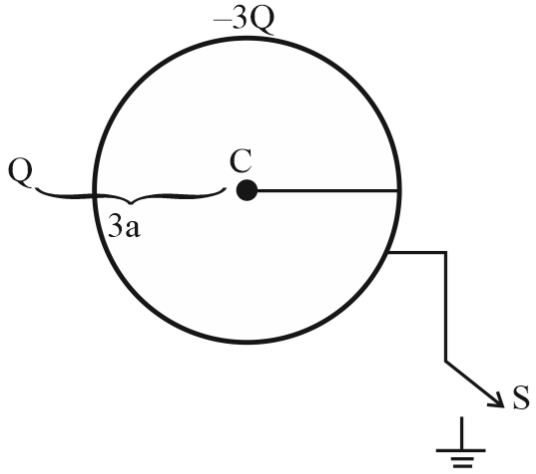
$r_2 = 2R$

Considering the circular motion of point mass m

$$M\omega_2^2 r_2 = \frac{GM^2}{36R^2}; \omega_2 = \sqrt{\frac{GM}{72R^3}} = \omega_1$$

8)

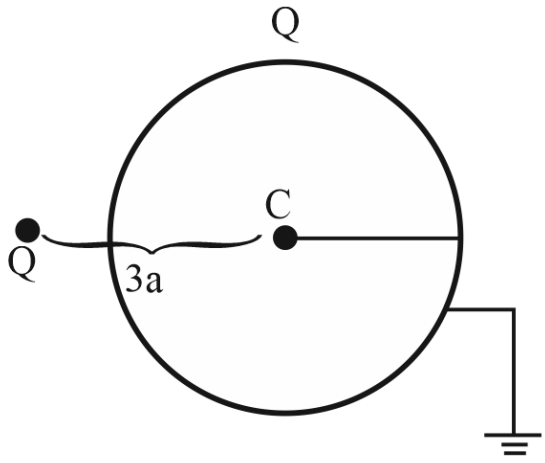
'S' i s open



$$\vec{E_C} = \vec{E}_{\text{due to O}} + \vec{E}_{\text{due to shell}}$$

$$\Rightarrow O = \frac{KQ}{9a^2}\hat{i} + \vec{E} \text{ due to shell}$$

$$V_C = \frac{KQ}{3a} + \frac{K(-3Q)}{a} = \frac{-8KQ}{3a}$$



$$V_C = O$$

$$\Rightarrow \frac{KQ}{3a} + \frac{Kq}{a} = 0$$

$$\Rightarrow q = -\frac{Q}{3}$$

'S' i s closed

9) External electric field and induced charge on outer surface of the sphere will produce a net zero field inside it. Net field at P must be vertically upward.

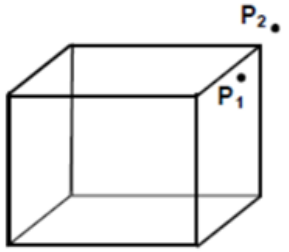
10) Using symmetry if charged particle lies at P_1 then $\phi_{OCDE} = \frac{q}{24\varepsilon_0}$

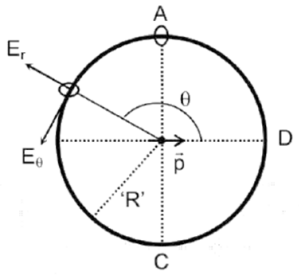
and $\phi_{OABC} = \frac{\left(\frac{q}{\varepsilon_0} - \frac{q}{8\varepsilon_0}\right)}{3} = \frac{7q}{24\varepsilon_0}$

If the charge particle lies at P_2 then

$\phi_{OCDE} = \frac{q}{24\varepsilon_0}$

and $\phi_{OABC} = \frac{q}{24\varepsilon_0}$





11)

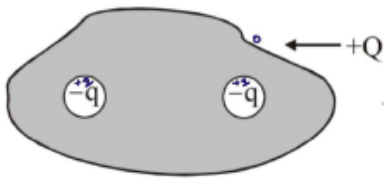
$E_r = \frac{2kp\cos\theta}{r^3}$

$E_\theta = \frac{kp\sin\theta}{r^3}$

From conservation of energy at A and E.

$0 + 0 = \frac{1}{2}mv^2 + \frac{(kp\cos\theta)q}{r^2}$

$v = \sqrt{\frac{-2kp(\cos\theta)q}{mr^2}} = \sqrt{\frac{-p(\cos\theta)q}{2\pi\varepsilon_0 mr^2}}$



12)

field outside zero potential decreases in the direction of field

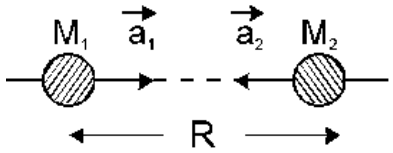
13) a + b = 7

a – b = 1

2a = 8
a = 4
b = 3
$R^2 = a^2 + b^2$
$R^2 = 4^2 + 3^2$
R = 5

14) $\vec{F}_1 + \vec{F}_2 = \vec{F}_3$
$\vec{F}_2 = \vec{F}_3 - \vec{F}_1$
$= (5\hat{i} - 4\hat{j}) - (2\hat{i} - 3\hat{j})$
$\vec{F}_2 = 3\hat{i} - \hat{j}$
$X = |\vec{F}_2| = \sqrt{10}$
$\frac{X}{\sqrt{10}} = 1$

15) Lets assume $\omega = 2\sigma^{\alpha}Q^{\beta}k^{\gamma}$.
Using dimensional analysis
$T^{-1} = (MT^{-2})\alpha\ (ML^{-3})^{\beta}\ (L^{-1})^{\gamma}$
On comparing
$0 = \alpha + \beta$,
$0 = -3\beta - \gamma$,
$-1 = -2\alpha$.
$\alpha = 1/2$, $\beta = -1/2$ and $\gamma = 3/2$
$\omega = 2\sqrt{\frac{\sigma k^3}{Q}}$
So, $v_g = \frac{\partial \omega}{\partial k} = 3\sqrt{\frac{\sigma k}{Q}}$
Substituting $k = 2\pi/\lambda$ and the rest of data given, we get
$v_g = 3\sqrt{\frac{2\pi\sigma}{Q\lambda}} = 0.6\ ms^{-1}$



16)
$a_1 = \frac{GM_1M_2}{R^2} / M_1$
$a_2 = \frac{GM_1M_2}{R^2} / M_2$
acceleation of M_1 w.r.t. M_2
$a_{rel.} = a_1 + a_2$
$= \frac{G(M_1 + M_2)}{R^2} = \frac{GM}{R^2}$

17) From property of ellipse

$a - ae = r_0$

$a + ae = r_1$

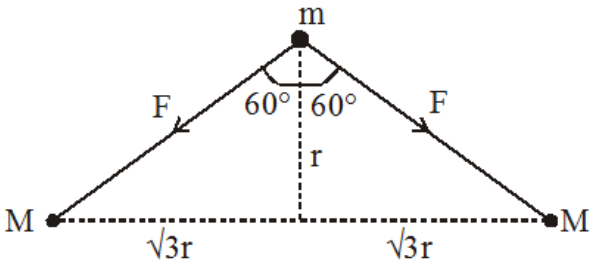
$$a = \frac{r_0 + r_1}{2}$$

$$e = \frac{r_1 - r_0}{r_1 + r_0}$$

Radius of curvature $= \frac{b^2}{a} = a^2 \times \frac{1 - e^2}{a} = \frac{2r_0 r_1}{r_0 + r_1}$

$\alpha = 6$

18)



$$2F\cos 60^\circ = 2\left[\frac{GMm}{(2r)^2}\right] \cdot \frac{1}{2} = \frac{mv^2}{r}$$

$$\Rightarrow V = \sqrt{\frac{GM}{4r}}$$

PART 2 : CHEMISTRY

19)

Milli-equivalent of NH_3 = mill. eq. of HNO_3(given) – m_{eq} of $HNO_{3(remaining)}$

$\therefore$ $m_{eq.}\text{of } HNO_{3(remaining)} = m_{eq.}\text{of } Ca(OH)_2$

$\Rightarrow$ $m_{eq.}$ of $HNO_{3(remaining)} = M \times V_{mL} \times n_f$

$m_{eq.}$ of $HNO_{3(remaining)} = 0.5 \times 30 \times 2$

$\therefore$ $m_{eq.}$ of $HNO_{3(remaining)} = 30$

$\therefore$ $m_{eq.}$ of $NH_3 = m_{eq.}$ of $HNO_{3(given)} - m_{eq.}$ of $HNO_{3(remaining)}$

$= 200 \times 0.25 \times 1 - 30$

$= 20$

$\therefore$ $\boxed{m_{eq.}\text{ of } NH_3 = 20}$

n_f of NH_3 in aq. solution $\Rightarrow (n_f = 1)$

$\therefore$ millimole $\times n_f = 20$ $\{n_f = 1\}$

$\therefore$ $m\,moles_{(NH_3)} = 20$

$N \rightarrow NH_3$

$\therefore$ m moles of N = m moles of NH_3

$\therefore$ mmoles of N = 20

$\therefore$ mass of $N = 20 \times 10^{-3} \times 14$ g

$\therefore$ Mass of N = 0.280 g

$\therefore$ $\%N = \frac{\text{Mass of N}}{\text{Mass of organic comp.}} \times 100$

$= \frac{0.280}{0.7} \times 100$
= 40%

20) $\text{Moles of Sugar} = \frac{1.71}{342} = 0.5 \times 10^{-2}$
$\text{Moles of C} = 12 \times 0.5 \times 10^{-2}$
No. of C atoms = $6 \times 10^{-2} \times 6 \times 10^{23}$
= 36×10^{21}
= 3.6×10^{22}

21) Milliequivalent of $KMnO_4$ = Milliequivalents of FeC_2O_4
$V(ml) \times 0.1 \times 5 = \frac{144}{144} \times 3$
V = 6 ml

22) $Cr_2O_7^{2\ominus} + 14H^{\oplus} + 6I^{\ominus} \rightarrow 3I_2 + 2Cr^{3\oplus} + 7H_2O$
(v.f.=3×2=6) (v.f.=2)
Equivalents of $K_2Cr_2O_7$ = equivalents of I_2
1 × 6 = moles of I_2 × 2
 Moles of I_2 = 3

23) O.N. of N is NH_4NO_3 is –3 & +5.
O.N. of S in FeS_2 = –1.
O.N. of F can be –1 or 0.
And alkaline earth metals can have only 0 & +2 as O.N.

24) Let $n_{Cu_2S} = a$; $n_{CuS} = a$
$Cu_2S + CuS + Ba(MnO_4)_2 + H^+ \rightarrow Cu^{2+} + SO_2 + Mn^{2+}$
n–factor = 8 6 10

25) Number of atoms in 1 gm–molecule depends on the nature of the molecule.

26) 6 Li (s) + N_2 (g) $\rightarrow$ $2Li_3N$(s)
Initial $\frac{8.4}{7}$ excess
moles = 1.2
so moles of N_2 consumed = $\frac{1}{6}' \times 1.2 = 0.2$
so volume of N_2 gas consumed at NTP = 0.2 × 22.4 = 4.48 litre
moles of Li3N produced = $\frac{2}{6} \times 1.2 = 0.4$
so moles of Li_3N = 0.4 × 35 = 14 gram
total atoms of product = 0.4 × 4 × N_A
ratio of moles reacted = 6 : 1

ratio of mass reacted = 42 : 28 = 3 : 2

27)

Avg. atomic mass = $\frac{(16 \times 90) + (10 \times 18)}{100}$

$= 16.2$

⇒ Atomic no. same for both so, Avg. protons per atom is 8.

⇒ Avg. no. of neutrons = Avg. atomic mass – Z = 8.2

28) Number of Hydrogen atom in 0.9 gm

$C_6H_{12}O_6 = \left(\frac{0.9}{180}N_A\right) 12 = \frac{3}{50}N_A$

Number of Hydrogen atom in 0.3 gm $C_2H_6 = \left(\frac{0.3}{30}N_A\right) 6 = \frac{3}{50}N_A$

29) With HPh only :

$(NV)_{HC\ell} = \frac{1}{2}(NV)_{Na_2CO_3}$

$0.05 \times x = \frac{0.05 \times 40 \times 2}{2}$

x = 40 ml

With MeOH only :

$(NV)_{HC\ell} = (NV)_{Na_2CO_3} + (NV)_{NaHCO_3}$

$0.05 \times y = (0.05 \times 40 \times 2) + (0.05 \times 40 \times 1)$

y = 120 ml

30) (A) B.R will be $2Zn + 5HNO_3 \rightarrow 2Zn(NO_3)_2 + \frac{1}{2}N_2O + \frac{5}{2}H_2O$

Electrons exchanged = 4

 n.f. = $\frac{4}{5}$

(B) O.N. of N increases & Cr decreases.

 Intramolecular redox.

(C) Balance reaction $(NH_4)_2Cr_2O_7 \rightarrow N_2 + Cr_2O_3 + 4H_2O$

Electron exchange = 6

 n.f. = 6.

(D) O.N. of Na increases.

 H_2 is oxidising agent.

31) POAC on S atom

$\frac{5.52 \times 10^3}{120} \times 2 = \frac{W}{98}$

 W = $\frac{5.52 \times 1000}{120} \times 2 \times 98$ = 9 Kg

32) $K_2Cr_2O_7 + 4H_2SO_4 + 3H_2S \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 3S + 7H_2O$

Every 294 g $K_2Cr_2O_7$ gives 96 g of S, So 3062.5 kg of $K_2Cr_2O_7$ is required to produce 1000 kg of sulphur.

33) Let divalent metal carbonate = MCO_3

$$36 = \frac{48}{(M + 60)} \times 100$$

$\Rightarrow M = 73.33$

Molar mass of metal carbonate = 73.33 + 60 = 133.33

Moles of metal carbonate = $\frac{266.66 \times 10^{-3}}{133.33} = 2 \times 10^{-3}$

Number of metal atoms = $2 \times 10^{-3} \times 6.02 \times 10^{23} = x \times 10^{20}$

x = 12.04

34) Let two equimolar carbonates are ACO_3 and BCO_3.

Wt. of metal A = 2.58 ×0.132 = 0.34 gm

$ACO_3 + BCO_3 \rightarrow AO + BO + 2CO_2$

Total two moles CO_3^{2-} provide 2 moles of CO_2

or 60 gm CO_3^{2-} provide 44 gm of CO_2

Mass of CO_3^{2-} required to produced 1.232 g of CO_2

$= \frac{60}{40} \times 1.232 = 1.68$ gm

so total wt. of carbonates = 1.68 gm

wt. of metal (A + B) = 2.58 – 1.68

= 0.90 gm

wt. of B = 0.90 – 0.34

= 0.56 gm ; % of B = $\frac{0.56}{2.58} \times 100 = 21.70$

35) $4Zn + NO_3^- + 7H_2O \rightarrow 4Zn^{2+} + NH_4^+ + 10OH^-$

36)

Let normality of $KMnO_4^-$ solution is N

$\therefore N \times 10 = 50 \times 1 \quad \Rightarrow N = 5$

molecular equation of MnO_4^- = molecular equation of oxalic acid

$$5 \times 100 = \frac{W}{E} \times 1000 \Rightarrow \left(\frac{W}{90/2}\right) \times 1000 \; W = 22.5 \text{ gm Ans.}$$

PART 3 : MATHEMATICS

37)

([x] – 5) ([x] – 3) + 2 < [x] – 5 + 2[x] – 6

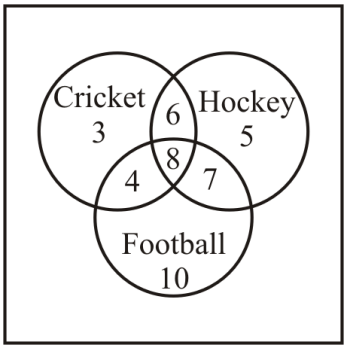
Let [x] = t

$\Rightarrow t^2 - 8t + 17 < 3t - 11$
$\Rightarrow t^2 - 11t + 28 < 0$
$\Rightarrow (t - 4)(t - 7) < 0$
$\Rightarrow 4 < [x] < 7$
$\Rightarrow 5 \le x < 7$
$\Rightarrow x = 5, 6$

38)

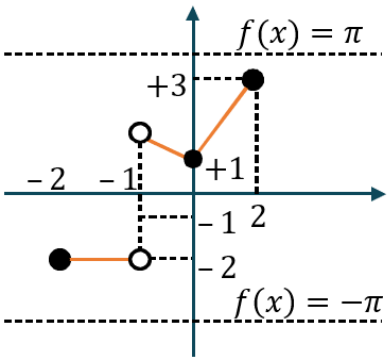
$f : R \to R$
$f(10 + x) = f(10 - x)$ $f(20 + x) = -f(20 - x)$
Let $10 - x = t$ $20 - x = a$
$f(t) = f(20 - t)$ $f(a) = -f(40 - a)$
$\Rightarrow$ Periodic with $f(a) = -f(-a)$
Period '40' $\Rightarrow$ odd function.

39) $f : N - \{1\} \to N$ $f(a) = \alpha$
Where α is max of powers of prime p such that p^{α} divides a. Also $g(a) = a + 1$
$\because$ $f(2) = 1$ $g(2) = 3$
$f(3) = 1$ $g(3) = 4$
$f(4) = 2$ $g(4) = 5$
$f(5) = 1$ $g(5) = 6$
$\Rightarrow f(2) + g(2) = 4$
$(f(3) + g(3)) = 5$
$f(4) + g(4) = 7$
$f(5) + g(5) = 7$
$\because$ Many one $f(x) + g(x)$ does not contain 1
$\Rightarrow$ into function
Ans. (D) [neither one-one nor onto]

40) Let C, H, F denote the sets of members who are on the cricket team, hockey team and football team respectively. 

Then we are given $n(C) = 21$, $n(H) = 26$, $n(F) = 29$
$n(H \cap C) = 14$, $n(H \cap F) = 15$, $n(F \cap C) = 12$
$n(C \cap H \cap F) = 8$
$n(C \cup H \cup F) = n(C) + (H) + n(F) - n(C \cap H) - n(H \cap F) - n(F \cap C) + n(C \cap H \cap F)$
Hence, $n(C \cup H \cup F) = (21 + 26 + 29) - (14 + 15 + 12) + 8 = 43$

41) $h(x) = g(f(x)) = \begin{cases} [f(x)], & -\pi \le f(x) < 0 \\ \sin(f(x)), & 0 \le f(x) \le \pi \end{cases}$



From graph of f(x), we get

$$\Rightarrow h(x) = \begin{cases} [[x]], & -2 \le x \le -1 \\ \sin(|x|+1), & -1 < x \le 2 \end{cases}$$

⇒ Domain of h(x) is [–2, 2]

and Range of h(x) is {–2, –1} ∪ [sin3, 1]

42)

f(2) = 3 f(1) ≠ 2

f(4) = 5 f(3) ≠ 4

so there are total 3 possibilities

Case-I : f={(1,1), (2,3), (3,2), (4,5), (5,4)}

Case-II : f={(1,4), (2,3), (3,1), (4,5),(5,2)}

Case-III : f = {(1,4), (2,3), (3,2), (4,5), (5,1)}

so $f^{-1}(2)$ can be 3 or 5

43) $2^{|x|+|y|} + 2^{|x|-|y|} = 2$

$$\Rightarrow 2^{|x|}\left(2^{|y|} + \frac{1}{2^{|y|}}\right) = 2$$

Here, $\left(2^{|y|} + \frac{1}{2^{|y|}}\right) \geqslant 2$

$$\Rightarrow \frac{2}{2^{|x|}} \geqslant 2 \Rightarrow 2 \geq 2 \times 2^{|x|}$$

$\Rightarrow 1 \geq 2^{|x|}$

$\Rightarrow 2^0 \geq 2^{|x|}$

$\Rightarrow |x| = 0$

$\Rightarrow x = 0$

Domain: = {0}

For x = 0;

$\Rightarrow 2^{0+|y|} + 2^{0-|y|} = 2$

$$\Rightarrow 2^{|y|} + \frac{1}{2^{|y|}} = 2$$

Hence, |y| = 0

⇒ y = 0

Range: {0}

Point (0, 0) lies on the line y = x

44)

Let $f(f(x)) = t$

$f(t) - t^3 = t$

$f(t) = t^3 + t$

number of real roots are $t^3 + t = 0$

$t(t^2+1) = 0$

$t = 0$

It is a polynomial function which is continuous everywhere.

(C) odd degree polynomial function is surjective for $R \to R$

(D) when $f(x) = f^{-1}(x)$

then $y = x$

$x^3 + x = x \Rightarrow x^3 = 0 \quad x = 0$

45) by $x = 7 \Rightarrow f(6) = 0$

by $x = 4 \Rightarrow f(4) = 0$

$\Rightarrow f(x) = (x - 4)(x - 6)\, g(x) \quad ...(1)$

(1) in given relation

$(x - 4)(x - 5)(x - 7)\, g(x - 1)$

$= (x - 7)(x - 4)(x - 6)\, g(x)$

$\Rightarrow (x - 5)\, g(x - 1) = (x - 6)\, g(x) \quad ...(2)$

by $x = 5 \Rightarrow g(5) = 0 \Rightarrow g(x) = (x - 5)\, h(x) \quad ...(3)$

(3) in (2) gives $(x - 5)(x - 6)\, h(x - 1)$

$= (x - 6)(x - 5)\, h(x)$

$\Rightarrow h(x) = h(x - 1) \Rightarrow h(x) = \lambda \quad ...(4)$

by (4), (3) in (1) $f(x) = (x - 4)(x - 6)(x - 5)\lambda$

by $f(7) = 6$, $\lambda = 1$

46) $x^8 - 22x^5 - 6x^3 + 3x^2 + 55 = 0$

x = even

Even – Even – Even + Even + odd ≠ 0

x = odd

Odd – Even – Even + Odd + Odd ≠ 0

number of solution = 0

47) Reflexive :-

Let $x \in R$

$\Rightarrow x \cdot x$ is an irrational number

$\Rightarrow x^2$ is irrational number which is not always true

take $x = \sqrt{2} \Rightarrow x^2 = 2$

it is not an irrational number

Symmetric :-

Take $(x, y) \in R$

$\Rightarrow x \cdot y$ is irrational number

$\Rightarrow y \cdot x$ is also an irrational number

$\Rightarrow (y, x) \in R$

 Relation is symmetric

Transitive :-

Take $x, y, z \in R$

$(x, y) \in R \Rightarrow xy$ is irrational number.

$\left(\sqrt{2},\sqrt{3}\right)\in R$ and $\left(\sqrt{3},2\sqrt{2}\right)\in R$ but $\left(\sqrt{2},2\sqrt{2}\right)\notin R$

Hence Relation is not transitive.

48) $\Rightarrow \alpha^{n+2}-\alpha^{n+1}-\alpha^{n}=0$ ....(1)

and $\beta^{n+2}-\beta^{n+1}-\beta^{n}=0$ ....(2)

$\alpha^{n+2}-\beta^{n+2}-(\alpha^{n+1}-\beta^{n+2})-(\alpha^{n}-\beta^{n})=0$

$$\Rightarrow \frac{\alpha^{n+2}-\beta^{n+2}}{\sqrt{5}}-\left(\frac{\alpha^{n+1}-\beta^{n+1}}{\sqrt{5}}\right)-\left(\frac{\alpha^{n}-\beta^{n}}{\sqrt{5}}\right)=0$$

$\Rightarrow T(n+2)-T(n-1)-T(n)=0$

$\Rightarrow T(n+2)=T(n+1)+T(n)$

(A) Put n = 2021

(B) $T_1=\dfrac{\alpha-\beta}{\sqrt{5}}=1$, $T_2=\dfrac{\alpha^2-\beta^2}{\sqrt{5}}=\dfrac{(\alpha-\beta)(\alpha+\beta)}{\sqrt{5}}=1$

$T_3=1+1=2$, $T_4=3$, $T_5=5$, $T_6=8$, $T_7=13$, $T_8=21$, $T_9=34$, $T_{10}=55$

(C) $$\sum_{n=1}^{2021}(T(n+2)-T(n+1))=\sum_{n=1}^{2021}T(n)$$

$$T(2023)-T(2)=\sum_{n=1}^{2021}T(n)$$

$$T(2023)-1=\sum_{n=1}^{2021}T(n)$$

49) f(x) is 1 now $g(f(x)<0 \Rightarrow g(f(1))<0$

50) $$g(x)=x^{2015}+\operatorname{sgn}x+\left[\frac{x^2+1}{p}\right]$$

for g(x) to be odd function $\left[\dfrac{x^2+1}{p}\right]=0$

i.e. $0\leqslant\dfrac{x^2+1}{p}<1$ $\Rightarrow$ p is positive

$x^2+1<p$

$p>5 \Rightarrow p=6$

51) $\because$ RHS = 1

$$\therefore \operatorname{sgn}\left(\left[\frac{15}{1+x^2}\right]\right)=1$$

$1+x^2\leq 15$

$\therefore x\in\{-3,-2,-1,0,1,2,3\}$

52) $x^3-x^2-x+1=(x+1)(x-1)^2$, f(x) = g(x) have 3 real and distinct roots

$\Rightarrow 0<a^2-2<1$

$\Rightarrow [a^2]=2$

Now $x^3-x^2-x+1=2$

has solution in (1,2)
⇒ the integral part of root is 1.

53)

$x^2 + xy + y^2 = 2$ ...(1)
$y^2 + yz + z^2 = 1$ ...(2)
$z^2 + xz + x^2 = 3$ ...(3)

$(1) - (2)$
$x^2 + xy + y^2 - y^2 - yz - z^2 = 1$
$xy - yz + x^2 - z^2 = 1$
$(x - z)(x + y + z) = 1$
$x + y + z = \frac{1}{x-z}$ ... (4)

$(3) - (1)$
$z^2 + xz + x^2 - x^2 - xy - y^2 = 1$
$x(z - y) + (z - y)(y + z) = 1$
$(z - y)(x + y + z) = 1$
$x + y + z = \frac{1}{z-y}$ ... (5)

By (4) and (5)
x – z = z – y
x + y = 2z ⇒ x, z, y are in A.P.
x = a – d, z = a, y = a + d
Putting these value in (1), (2) and (3)
$(a - d)^2 + a^2 - d^2 + (a + d)^2 = 2$
$2(a^2 + d^2) + a^2 - d^2 = 2$
$3a^2 + d^2 = 2$ ...(6)
By equation (2)
$(a + d)^2 + a(a + d) + a^2 = 1$
$3a^2 + d^2 + 3ad = 1$ ...(7)
By equation (3)
$a^2 + a^2 - ad + a^2 + d^2 - 2ad = 3$
$3a^2 + d^2 - 3ad = 3$ ...(8)
By (7) – (8)
6ad = –2
$ad = -\frac{1}{3} \Rightarrow d = -\frac{1}{3a}$ ...(9)
By (6) and (9)
$3a^2 + \frac{1}{9a^2} = 2$
$27a^4 - 18a^2 + 1 = 0$
$a^2 = \frac{18 \pm 6\sqrt{6}}{27 \times 2}$
$= \frac{3 \pm \sqrt{6}}{9}$
$a^2 = \frac{3 + \sqrt{6}}{9}$ $d^2 = \frac{1}{3 + \sqrt{6}} = \frac{3 - \sqrt{6}}{3}$
xy + yz + zx = (a – d) (a + d) + (a + d) a + (a) (a – d)
$= a^2 - d^2 + a^2 + ad + a^2 - ad$
$= 3a^2 - d^2$
Substituting value of a^2 and d^2 →
$= \frac{3 + \sqrt{6}}{3} - \left(\frac{3 - \sqrt{6}}{3}\right) = \frac{2\sqrt{6}}{3} = \sqrt{\frac{4 \times 6}{9}}$

$= \sqrt{\frac{8}{3}} = \sqrt{\frac{\alpha}{\beta}}$

$\Rightarrow \quad \alpha = 8, \beta = 3$

$\alpha - \beta = 5$

54) Number of '–1' in the series = a
Number of '1' in the series = b
Number of '2' in the series = c

$-a + b + c = 19 \qquad ...(i)$

$a + b + 4c = 99 \qquad ...(ii)$

(i) + (ii)

$2b + 5c = 118$

$b = \frac{118 - 5c}{2}$

$a = b + c - 19$

$= \frac{118 - 5c}{2} + c - 19$

$= \frac{80 - 3c}{2}$

$\Sigma x_i^3 \Big)_{\substack{c=0 \\ a=40 \\ b=59}} = 19$

for minimum of